Question 1:

Write a C program to find the curved surface area and volume of a cylinder when radius is given in millimeters and height is given in centimeters.

1) **Understanding the problem statements**

The problem requires calculating two properties of a cylinder:

- **Curved Surface Area (CSA)**
- **Volume**

The radius and height are provided in different units:

- Radius → **millimeters (mm)**
- Height → **centimeters (cm)**

Therefore, before performing the calculations, both dimensions must be converted into the same unit. Since the height is already given in centimeters, the radius will be converted from millimeters to centimeters.

Required Input(s)

- Radius of the cylinder in millimeters
- Height of the cylinder in centimeters

Required Output(s)

- Curved Surface Area in square centimeters (cm^2)
- Volume in cubic centimeters (cm^3)

Unit Conversion

1 cm = 10 mm

Therefore:

Radius in cm = Radius in mm / 10

Formula Used

Let:

- r = Radius in centimeters
- h = Height in centimeters

Curved Surface Area:

CSA = $2\pi rh$

Volume:

V = $\pi r^2 h$

Where $\pi = 3.14159$.

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2. Standard Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	Radius = 10 mm, Height = 10 cm	CSA = 62.8318 cm^2, Volume = 31.4159 cm^3	CSA = 62.8318 cm^2, Volume = 31.4159 cm^3	Pass
2	Radius = 20 mm, Height = 15 cm	CSA = 188.4954 cm^2, Volume = 188.4954 cm^3	CSA = 188.4954 cm^2, Volume = 188.4954 cm^3	Pass
3	Radius = 50 mm, Height = 20 cm	CSA = 628.3180 cm^2, Volume = 1570.7950 cm^3	CSA = 628.3180 cm^2, Volume = 1570.7950 cm^3	Pass
4	Radius = 25 mm, Height = 12 cm	CSA = 188.4954 cm^2, Volume = 235.6193 cm^3	CSA = 188.4954 cm^2, Volume = 235.6193 cm^3	Pass
5	Radius = 100 mm, Height = 30 cm	CSA = 1884.9540 cm^2, Volume = 9424.7700 cm^3	CSA = 1884.9540 cm^2, Volume = 9424.7700 cm^3	Pass

3. Algorithm

Algorithm: To Find the Curved Surface Area and Volume of a Cylinder

1. Start.
2. Declare variables for radius in millimeters, radius in centimeters, height, curved surface area, and volume.
3. Assign the value of π as 3.14159.
4. Input the radius of the cylinder in millimeters.
5. Input the height of the cylinder in centimeters.
6. Convert the radius from millimeters to centimeters by dividing it by 10.
7. Calculate the curved surface area using the formula:

CSA = 2 × π × r × h

8. Calculate the volume using the formula:

V = π × r × r × h

9. Display the curved surface area in square centimeters.
10. Display the volume in cubic centimeters.
11. Stop.

Evaluation of the Algorithm Against Test Cases

The algorithm was evaluated using the test cases prepared above.

For each test case:

- The radius was first converted from millimeters to centimeters.
- The converted radius and given height were used in the CSA formula.
- The same values were used to calculate the volume.
- The calculated results matched the expected outputs.

For example, in Test Case 1:

- Radius = 10 mm = 1 cm

- Height = 10 cm

CSA:

$2 \times 3.14159 \times 1 \times 10 = 62.8318\ cm^2$

Volume:

$3.14159 \times 1 \times 1 \times 10 = 31.4159\ cm^3$

The result matches the expected output. Similarly, the algorithm produces correct results for all the remaining test cases.

Therefore, **the algorithm successfully passes all the prepared test cases**.

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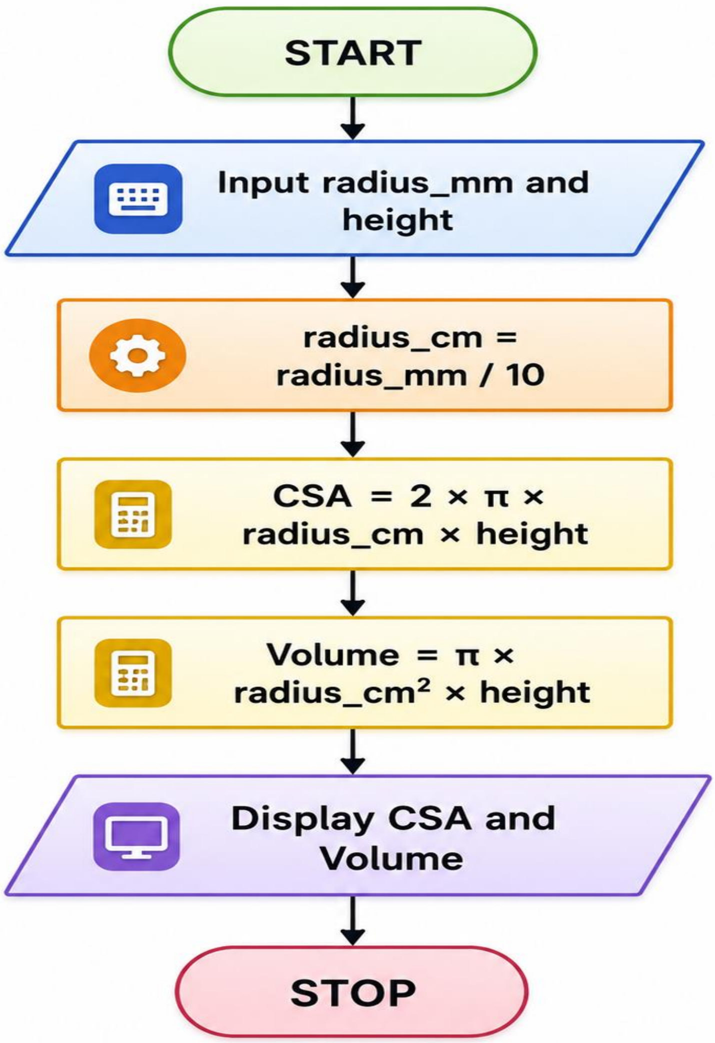
4. Flowchart



Flowchart Symbols to Use While Drawing

Step	Symbol
Start/Stop	Oval
Input/Output	Parallelogram
Calculations	Rectangle
Direction	Arrow

Evaluation of the Flowchart Against Test Cases

The flowchart follows the same logical sequence as the algorithm.

For every test case:

1. It accepts the radius and height.
2. It converts the radius from millimeters to centimeters.
3. It calculates the curved surface area.
4. It calculates the volume.
5. It displays both results.

Using Test Case 1 as an example, the flowchart converts 10 mm into 1 cm and correctly calculates:

- CSA = 62.8318 cm^2
- Volume = 31.4159 cm^3

The same sequence works correctly for all the other test cases.

Therefore, **the flowchart successfully passes all the prepared test cases**.

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5. C Program

```
#include <stdio.h>

int main()
```

```
{
    float radius_mm, radius_cm, height;
    float curved_surface_area, volume;
    const float PI = 3.14159;

    printf("Enter the radius in millimeters: ");
    scanf("%f", &radius_mm);

    printf("Enter the height in centimeters: ");
    scanf("%f", &height);

    radius_cm = radius_mm / 10;

    curved_surface_area = 2 * PI * radius_cm * height;

    volume = PI * radius_cm * radius_cm * height;

    printf("\nRadius in centimeters = %.2f cm", radius_cm);
    printf("\nCurved Surface Area = %.4f square centimeters",
curved_surface_area);
    printf("\nVolume = %.4f cubic centimeters\n", volume);

    return 0;
}
```

6. Compilation and Execution

Save the program with the filename using gedit cylinder.c

Compilation Command

cc cylinder.c

If there are no compilation errors, execute the program using:

./a.out

Sample Execution

Test Case

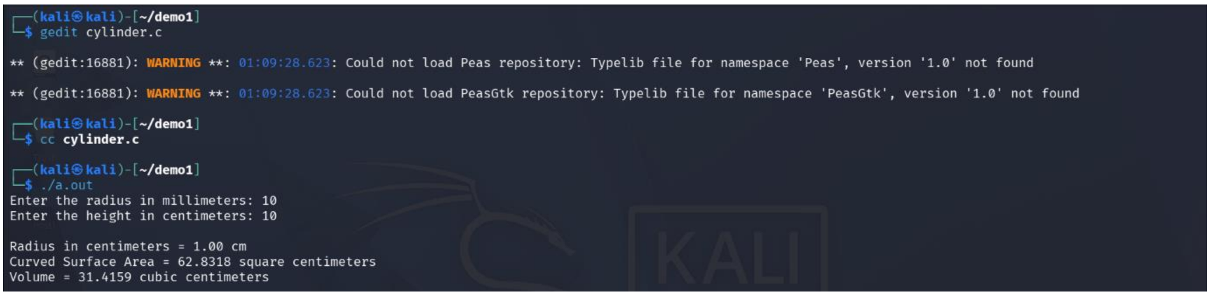
Enter the radius in millimeters: 10

Enter the height in centimeters: 10

Radius in centimeters = 1.00 cm

Curved Surface Area = 62.8318 square centimeters

Volume = 31.4159 cubic centimeters



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7. Evaluation of the Code Against Test Cases

After compiling and executing the program, the code was tested using all the test cases prepared in Step 2.

Test Case	Expected Result	Actual Result	Status
1	Correct CSA and Volume	Matched	Pass

Test Case	Expected Result	Actual Result	Status
2	Correct CSA and Volume	Matched	Pass
3	Correct CSA and Volume	Matched	Pass
4	Correct CSA and Volume	Matched	Pass
5	Correct CSA and Volume	Matched	Pass

The program correctly converts the radius from millimeters to centimeters before performing calculations. The calculated curved surface area and volume match the expected outputs for all test cases.

Therefore, **the code successfully passes all the prepared test cases**.

Final Result

Thus, the C program to find the curved surface area and volume of a cylinder when the radius is given in millimeters and height is given in centimeters was successfully designed, implemented, compiled, and executed. The algorithm, flowchart, and program were evaluated using multiple test cases, and all test cases passed successfully.

Question 2:

Write a C program to find the remaining area when a triangle is removed from a rectangle.

1. Understanding the Problem Statement

The problem requires calculating the **remaining area of a rectangle after removing a triangular portion from it**.

First, the area of the rectangle is calculated. Then, the area of the triangle is calculated and subtracted from the rectangle's area.

Required Input(s)

- Length of the rectangle
- Breadth of the rectangle

- Base of the triangle
- Height of the triangle

Required Output(s)

- Area of the rectangle
- Area of the triangle
- Remaining area after removing the triangle

Formula Used

Let:

- l = Length of rectangle
- b = Breadth of rectangle
- base = Base of triangle
- height = Height of triangle

Area of Rectangle:

Area = Length × Breadth

Area of Triangle:

Area = ½ × Base × Height

Remaining Area:

Remaining Area = Area of Rectangle − Area of Triangle

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2. Standard Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	Length = 10 cm, Breadth = 8 cm, Base = 6 cm, Height = 4 cm	Rectangle Area = 80 cm^2, Triangle Area = 12 cm^2, Remaining Area = 68 cm^2	Rectangle Area = 80 cm^2, Triangle Area = 12 cm^2, Remaining Area = 68 cm^2	Pass

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
2	Length = 20 cm, Breadth = 10 cm, Base = 10 cm, Height = 6 cm	Rectangle Area = 200 cm^2, Triangle Area = 30 cm^2, Remaining Area = 170 cm^2	Rectangle Area = 200 cm^2, Triangle Area = 30 cm^2, Remaining Area = 170 cm^2	Pass
3	Length = 15 cm, Breadth = 12 cm, Base = 8 cm, Height = 5 cm	Rectangle Area = 180 cm^2, Triangle Area = 20 cm^2, Remaining Area = 160 cm^2	Rectangle Area = 180 cm^2, Triangle Area = 20 cm^2, Remaining Area = 160 cm^2	Pass
4	Length = 25 cm, Breadth = 20 cm, Base = 12 cm, Height = 10 cm	Rectangle Area = 500 cm^2, Triangle Area = 60 cm^2, Remaining Area = 440 cm^2	Rectangle Area = 500 cm^2, Triangle Area = 60 cm^2, Remaining Area = 440 cm^2	Pass
5	Length = 30 cm, Breadth = 15 cm, Base = 20 cm, Height = 8 cm	Rectangle Area = 450 cm^2, Triangle Area = 80 cm^2, Remaining Area = 370 cm^2	Rectangle Area = 450 cm^2, Triangle Area = 80 cm^2, Remaining Area = 370 cm^2	Pass

3. Algorithm

Algorithm: To Find the Remaining Area When a Triangle is Removed from a Rectangle

1. Start.
2. Declare variables for length and breadth of the rectangle.
3. Declare variables for base and height of the triangle.
4. Declare variables for rectangle area, triangle area, and remaining area.
5. Input the length and breadth of the rectangle.

6. Input the base and height of the triangle.
7. Calculate the area of the rectangle using the formula:

Rectangle Area = Length × Breadth

8. Calculate the area of the triangle using the formula:

Triangle Area = ½ × Base × Height

9. Calculate the remaining area by subtracting the triangle area from the rectangle area.
10. Display the rectangle area.
11. Display the triangle area.
12. Display the remaining area.
13. Stop.

Evaluation of the Algorithm Against Test Cases

The algorithm was evaluated using the test cases prepared above.

For each test case:

- The length and breadth were used to calculate the area of the rectangle.
- The base and height were used to calculate the area of the triangle.
- The triangle area was subtracted from the rectangle area to obtain the remaining area.

For example, in Test Case 1:

- Length = 10 cm
- Breadth = 8 cm
- Base = 6 cm
- Height = 4 cm

Rectangle Area:

$10 \times 8 = 80$ cm^2

Triangle Area:

$\frac{1}{2} \times 6 \times 4 = 12\ cm^2$

Remaining Area:

$80 - 12 = 68\ cm^2$

The result matches the expected output. Similarly, the algorithm produces correct results for all the remaining test cases.

Therefore, **the algorithm successfully passes all the prepared test cases**.

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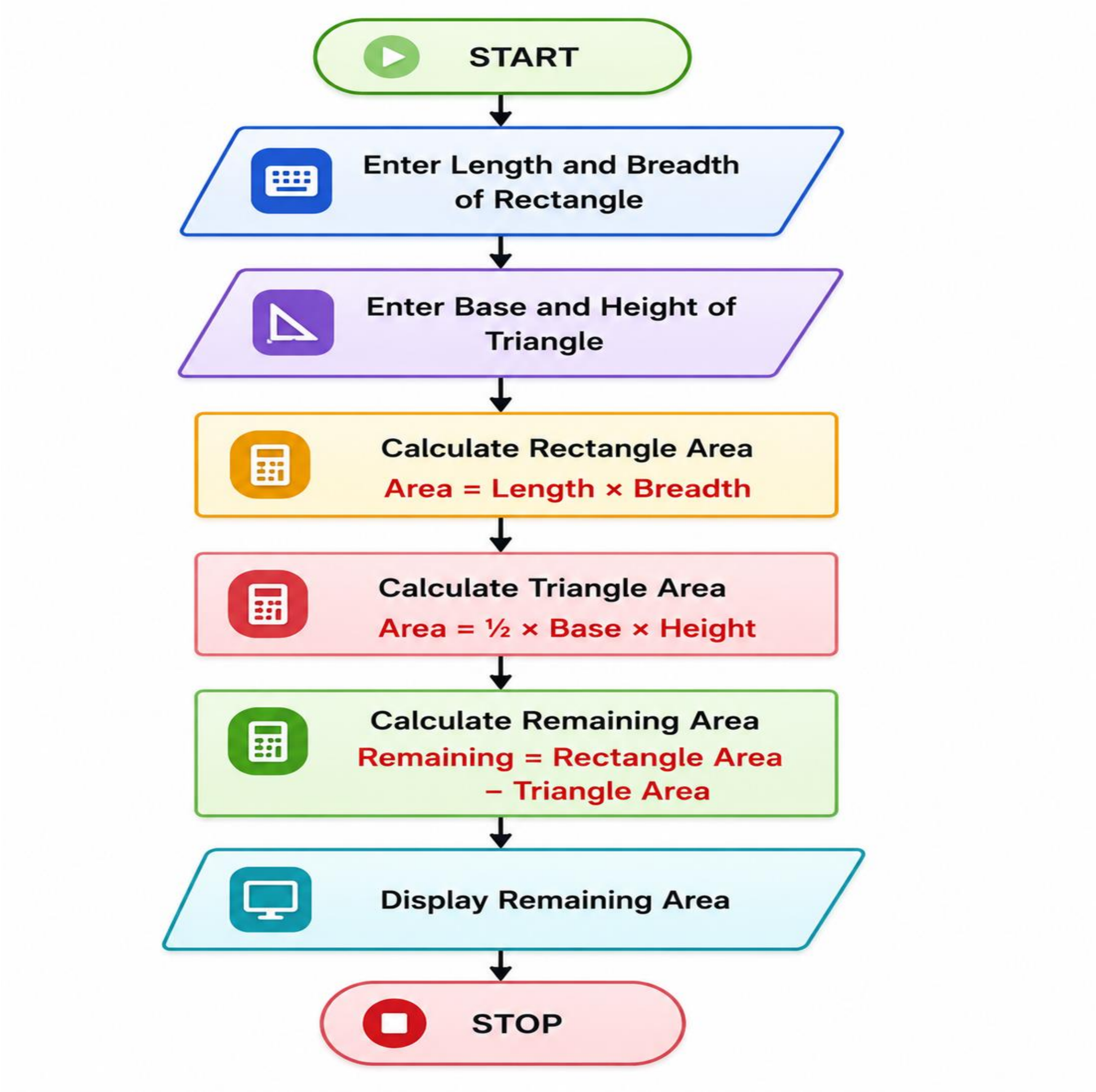
4. Flowchart



Evaluation of the Flowchart Against Test Cases

The flowchart follows the same logical sequence as the algorithm.

For every test case:

1. It accepts the dimensions of the rectangle.
2. It accepts the base and height of the triangle.
3. It calculates the rectangle area.
4. It calculates the triangle area.
5. It subtracts the triangle area from the rectangle area.
6. It displays the remaining area.

Using Test Case 1 as an example:

- Rectangle Area = 80 cm^2
- Triangle Area = 12 cm^2
- Remaining Area = 68 cm^2

The same sequence works correctly for all the other test cases.

Therefore, **the flowchart successfully passes all the prepared test cases**.

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5. C Program

```
#include <stdio.h>

int main()

{

   float length, breadth, base, height;

   float rectangle_area, triangle_area, remaining_area;

   printf("Enter the length of the rectangle: ");

   scanf("%f", &length);

   printf("Enter the breadth of the rectangle: ");
```

```
   scanf("%f", &breadth);

   printf("Enter the base of the triangle: ");

   scanf("%f", &base);

   printf("Enter the height of the triangle: ");

   scanf("%f", &height);

   rectangle_area = length * breadth;

   triangle_area = 0.5 * base * height;

 remaining_area = rectangle_area - triangle_area;

   printf("\nArea of Rectangle = %.2f square centimeters", rectangle_area);

   printf("\nArea of Triangle = %.2f square centimeters", triangle_area);

   printf("\nRemaining Area = %.2f square centimeters\n", remaining_area);

   return 0;

}
```

```
#include <stdio.h>

int main()
{
    float length, breadth, base, height;
    float rectangle_area, triangle_area, remaining_area;

    printf("Enter the length of the rectangle: ");
    scanf("%f", &length);

    printf("Enter the breadth of the rectangle: ");
    scanf("%f", &breadth);

    printf("Enter the base of the triangle: ");
    scanf("%f", &base);

    printf("Enter the height of the triangle: ");
    scanf("%f", &height);

    rectangle_area = length * breadth;

    triangle_area = 0.5 * base * height;

    remaining_area = rectangle_area - triangle_area;

    printf("\nArea of Rectangle = %.2f square centimeters", rectangle_area);
    printf("\nArea of Triangle = %.2f square centimeters", triangle_area);
    printf("\nRemaining Area = %.2f square centimeters\n", remaining_area);

    return 0;
}
```

6. Compilation and Execution

Save the program with the filename:

remaining_area1.c using gedit

Compilation Command

cc remaining_area1.c

If there are no compilation errors, execute the program using:

./a.out

```
┌──(kali㉿kali)-[~/demo1]
└─$ cc area1.c

┌──(kali㉿kali)-[~/demo1]
└─$ ./a.out
Enter the length of the rectangle: 10
Enter the breadth of the rectangle: 8
Enter the base of the triangle: 6
Enter the height of the triangle: 4

Area of Rectangle = 80.00 square centimeters
Area of Triangle = 12.00 square centimeters
Remaining Area = 68.00 square centimeters
```

Sample Execution

Test Case

Enter the length of the rectangle: 10

Enter the breadth of the rectangle: 8

Enter the base of the triangle: 6

Enter the height of the triangle: 4

Area of Rectangle = 80.00 square centimeters

Area of Triangle = 12.00 square centimeters

Remaining Area = 68.00 square centimeters

<u>7. Evaluation of the Code Against Test Cases</u>

After compiling and executing the program, the code was tested using all the test cases prepared in Step 2.

Test Case	Expected Result	Actual Result	Status
1	Remaining Area = 68 cm^2	Matched	Pass
2	Remaining Area = 170 cm^2	Matched	Pass
3	Remaining Area = 160 cm^2	Matched	Pass

Test Case	Expected Result	Actual Result	Status
4	Remaining Area = 440 cm^2	Matched	Pass
5	Remaining Area = 370 cm^2	Matched	Pass

The program correctly calculates the area of both shapes and subtracts the triangle's area from the rectangle's area. The calculated outputs match the expected results for all test cases.

Therefore, **the code successfully passes all the prepared test cases**.

Final Result

Thus, the C program to find the remaining area when a triangle is removed from a rectangle was successfully designed, implemented, compiled, and executed. The algorithm, flowchart, and program were evaluated using multiple test cases, and all test cases passed successfully.